history of machine learning, where an agent is self-deciding and autonomous.

Figure 1.1 shows four cases where a user may interact with a large language model (LLM) directly or through an agent/assistant proxy, an agent/assistant, or an autonomous agent. These four use cases are highlighted in more detail in this list:

- *Direct user interaction*—If you used earlier versions of ChatGPT, you experienced direct interaction with the LLM. There is no proxy agent or other assistant interjecting on your behalf.
- *Agent/assistant proxy*—If you've used Dall-E 3 through ChatGPT, then you've experienced a proxy agent interaction. In this use case, an LLM interjects your requests and reformulates them in a format better designed for the task. For example, for image generation, ChatGPT better formulates the prompt. A proxy agent is an everyday use case to assist users with unfamiliar tasks or models.
- *Agent/assistant*—If you've ever used a ChatGPT plugin or GPT assistant, then you've experienced this use case. In this case, the LLM is aware of the plugin or assistant functions and prepares to make calls to this plugin/function. However, before making a call, the LLM requires user approval. If approved, the plugin or function is executed, and the results are returned to the LLM. The LLM then wraps this response in natural language and returns it to the user.
- *Autonomous agent*—In this use case, the agent interprets the user's request, constructs a plan, and identifies decision points. From this, it executes the steps in the plan and makes the required decisions independently. The agent may request user feedback after certain milestone tasks, but it's often given free rein to explore and learn if possible. This agent poses the most ethical and safety concerns, which we'll explore later.